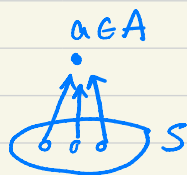


A : set

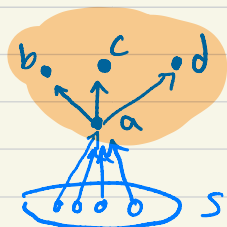
R : partial order on A

S : subset of A

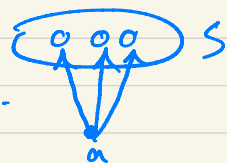
- If all elements of S are related to some $a \in A$ then a is an upper bound (ub) for S .



- If set a is an upper bound for S and is related to all upper bounds of S , then a is called the least upper bound (lub) for S .

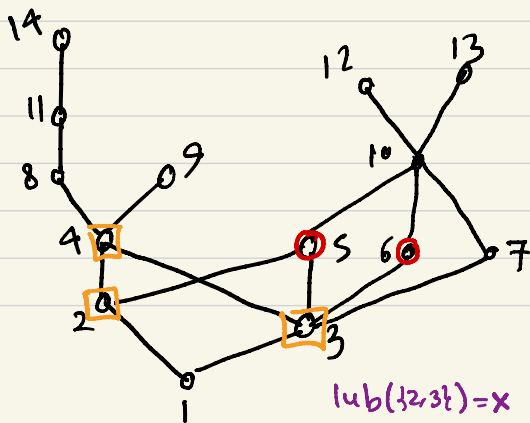
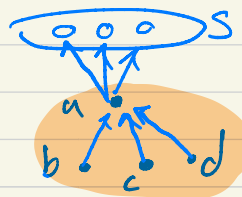


- If element $a \in A$ is related to all elements in S then a is a lower bound (lb) for S .



- If set a is lower bound for S and all other lower bounds of S are related to a , then

element a is called the greatest lower bound (glb) for S .



$$S = \{5, 6\}$$

$$T = \{2, 3, 4\}$$

$$ub(S) = \{10, 12, 13\}$$

$$ub(T) = \{9, 8, 11, 14\}$$

$$lub(S) = 10$$

$$lub(T) = 4$$

$$lb(S) = \{3, 1\}$$

$$lb(T) = \{1\}$$

$$glb(S) = 3$$

$$glb(T) = 1$$

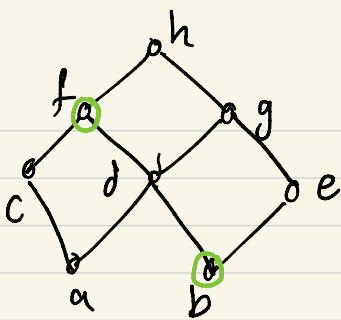
$$glb(\{3, 4, 5\}) = 3$$

$$glb(\{4, 5, 6, 7\}) = ?$$

$$lub(\{3, 4, 5\}) = X$$

$$lub(\{4, 5, 6, 7\}) = ?$$

$$lub(\{2, 3\}) = X$$



$$S = \{b, f\}$$

$$\text{lub}(S) = \{h, f\}$$

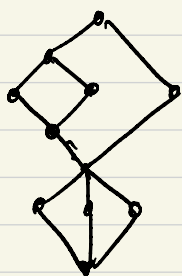
$$\text{lub}(S) = f$$

$$\text{lub}(S) = \{b\}$$

$$\text{glb}(S) = b$$

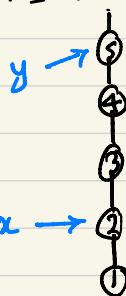
lattice: a poset where all pairs have glb and lub.

$\text{glb}(a, b)$ does not exist, therefore this is not a lattice.



this
is
a
lattice

• Is $(\leq, \mathcal{N})$ a lattice? **yes**



$$\text{lub}(x, y) = \max\{x, y\}$$

$$\text{glb}(x, y) = \min\{x, y\}$$

• Let A be a set. Is $(\subseteq, 2^A)$ a lattice?

$$x, y \subseteq A$$

$$\text{lub}(x, y) = x \cup y$$

$$\text{glb}(x, y) = x \cap y$$

verify this on picture from previous lecture.

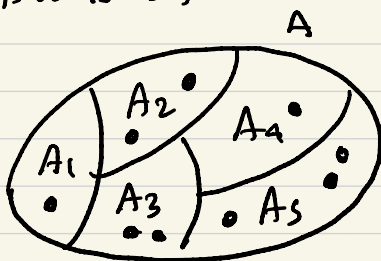
A partition of a set A is a collection of nonempty disjoint subsets of A s.t. their union is A.

a collection $A_1, A_2, \dots, A_k$ of subsets of A s.t.:

① $A_i \neq \{\}$

② $A_i \cap A_j = \{\}$

③ $A_1 \cup A_2 \cup \dots \cup A_k = A$



exmp: $A = \{1, 2, 3, 4, 5, 6\}$

$$A_1 = \{1, 2, 3\}$$

$$A_2 = \{5\}$$

$$A_3 = \{4, 6\}$$

A_1, A_2, A_3 is a partition of A

$$A_1 = \{1\}$$

$$A_2 = \{2, 6\}$$

$$A_3 = \{5\}$$

$$A_4 = \{3, 4\}$$

A_1, A_2, A_3, A_4 is a partition of A

exmp: $A = \{a, b, c, d, e\}$

$$R = \{(\underline{a}, a), (a, c), (a, e), (\underline{b}, b), (b, d), (c, a), (\underline{c}, c), (c, e), (d, b), (\underline{d}, d), (e, a), (e, c), (\underline{e}, e)\}$$

reflexive, symmetric, transitive $\Rightarrow R$ is an equivalence relation

for each $x \in A$ we define
equivalence class of x as:

$$[x] = \{y \in A : x R y\}$$

$$[a] = \{a, e, c\}$$

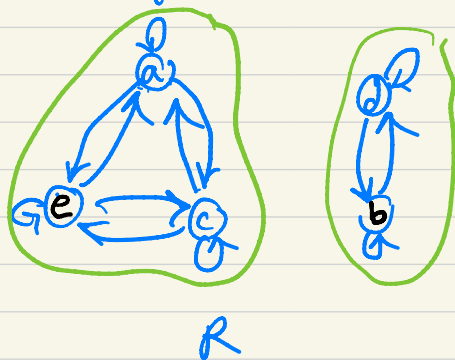
$$[c] = \{a, e, c\}$$

$$\Rightarrow [a] = [c] = [e]$$

$$[d] = \{d, b\}$$

$$[e] = \{d, b\}$$

$$[d] = [b]$$



Relation R partitions the set A into two
equivalent classes $\{a, e, c\}, \{d, b\}$

Any equivalence relation partitions the ground set
into same equivalence classes.

exmp: $A = \{1, 2, 3, 4, 5, 6, 7\}$

$$x R y: 3 \mid x - y$$

H.W. Prove that R is an equivalence relation.

Identify equivalence classes of R .

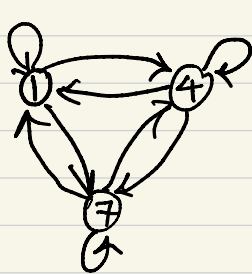
$$4 R 1: 3 \mid 4 - 1 \checkmark$$

$$2 R 5: 3 \mid 2 - 5 \checkmark$$

$$7 R 1: 3 \mid 7 - 1 \checkmark$$

$$~~2 R 7: 3 \mid 2 - 7 \times~~$$

$$1 R 7: 3 \mid 1 - 7 \checkmark$$



$$[1] = \{1, 4, 7\} = [4] = [7]$$

$$[2] = \{2, 5\}$$

$$[3] = \{3, 6\}$$

$$A = [1] \cup [2] \cup [3]$$

exmp: $(\mathbb{R}, \mathbb{Z})$

$$a R b : a \bmod 4 = b \bmod 4$$

↑
remainder of division

$$\begin{array}{l} 13 \bmod 4 = 1 \\ 21 \bmod 4 = 1 \end{array} \left\{ \begin{array}{l} 13 R 21, 21 R 13 \end{array} \right.$$

$$\begin{array}{l} 2 \bmod 4 = 2 \\ 14 \bmod 4 = 2 \end{array} \left\{ \begin{array}{l} 2 R 14, 14 R 2 \end{array} \right.$$

H.W. verify R is an equivalence relation

- reflexive $\forall x \in \mathbb{Z} : x \bmod 4 = x \bmod 4 \Rightarrow x R x$
- symmetric
- transitive

R partitions $\mathbb{Z}$ into equivalence classes. identify classes?

for any $x \in \mathbb{Z}$ we have $x \bmod 4 \in \{0, 1, 2, 3\}$

$$[0] = \{x \in \mathbb{Z} : x \bmod 4 = 0\} = \{\dots, -4, 0, 4, 8, 12, \dots\}$$

$$[1] = \{x \in \mathbb{Z} : x \bmod 4 = 1\} = \{\dots, -3, 1, 5, 9, \dots\}$$

$$[2] = \{x \in \mathbb{Z} : x \bmod 4 = 2\} = \{\dots, -2, 2, 6, 10, \dots\}$$

$$[3] = \{x \in \mathbb{Z} : x \bmod 4 = 3\} = \{\dots, -1, 3, 7, \dots\}$$

$$\begin{aligned} \mathbb{Z} &= [0] \cup [1] \cup [2] \cup [3] \\ &= [0] \cup [5] \cup [-2] \cup [3] \end{aligned}$$